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The Modak–Walawalkar Framework

Think Manifolds, Not PDEs

Physics described as Bayesian priors. The state space cut from a hypercube to a learned Riemannian manifold. One geometric object that is both the physical law and the measure of risk, carried across eight domains by changing only the priors.

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The main innovation, in brief

• **A new object for physics.** For 339 years, computational physics has run one pipeline: write a Lagrangian, derive the PDEs, solve them. Every modern advance, from supercomputers to PINNs, makes the *solve* faster. MW changes the *object*. The law of a physical domain is represented as a learned Riemannian manifold of admissible states.

• **Priors as the language of physics.** Physical knowledge is written deliberately as Bayesian priors: bounded quantities as Beta, limits as truncated distributions, rate constants as LogNormal, regime switches as Bernoulli, quantum measurement as heavy-tailed Student-t. The priors are the specification. No PDE is derived.

• **State-space reduction.** The priors cut the ambient hypercube, which Monte Carlo and grid methods must sample corner to corner, down to the thin set of physically realisable states. A VAE learns that set as a manifold with a metric $G(z) = J^{T}WJ$. All inference then runs on the manifold, not in the hypercube. For 1,000 qubits this is a 2,000-dimensional fibre bundle instead of a 2^{1000}-dimensional state vector.

• **One object, two structures.** The manifold *is* the model. The distance to it *is* the risk. On the manifold the MW distance coincides with Synge's world function; off it, the MW distance extends Synge into a physics-grounded anomaly score that needs no labelled failures.

• **Carried across domains.** One engine with the same encode–decode–distance skeleton, now applied to batteries, RF spectrum and electronic warfare, cyber-physical security, quantum computing, quantum chemistry, fluid dynamics, gravitational physics and quantum gravity. Each port changes only the priors.

• **Physics combined by adding priors.** Each body of physics is a set of priors, so each defines a submanifold. To combine two theories, put both prior sets on one manifold and learn their intersection. Quantum mechanics and general relativity, kept apart for a century by incompatible PDEs, are held by one 20-dimensional MW object.

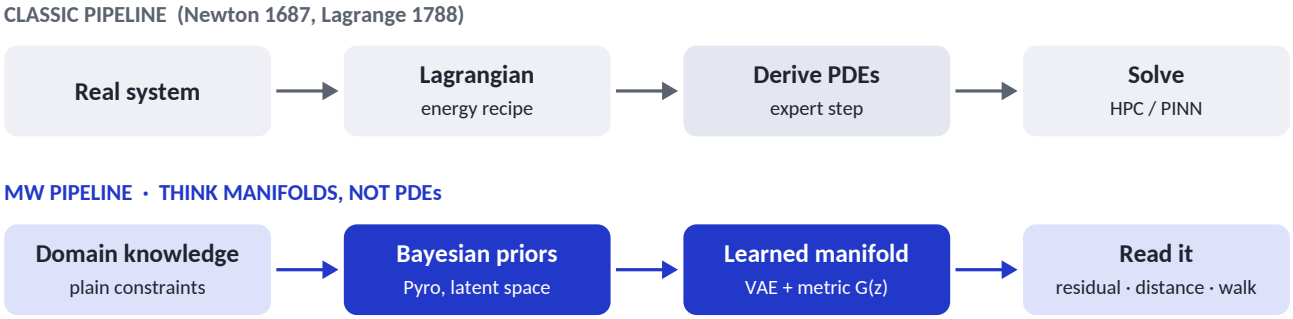
1 · A pipeline nobody questioned for 339 years

Newton's *Principia* (1687) and Lagrange's mechanics (1788) fixed the way physics is computed. Describe the system as a Lagrangian. Run it through the Euler–Lagrange equations. Out come the governing PDEs: Maxwell, Navier–Stokes, Schrödinger, Einstein. Then solve them.

Everything since has accelerated the last step. Finite elements, spectral methods, GPUs, neural operators and physics-informed neural networks all assume the PDE is already in hand. The derivation step still demands tensor calculus and differential geometry. Most engineers who understand their system cannot clear that bar. And for fleets of batteries, crowded RF spectra or live network telemetry, re-solving equations as conditions change is too slow.

Those who step outside the PDE pipeline, through Monte Carlo, grids or black-box machine learning, still work in the full ambient space. They sample the whole hypercube of possible inputs, almost all of which no physical system can reach. The cost grows exponentially with dimension. The physics is used, at best, as a penalty after the fact.

2 · The main innovation: think manifolds, not PDEs



The derivation step is replaced by a specification step. Equations, if wanted, come out as outputs.

Figure 1. The derivation step is replaced by a specification step. The expert writes what is physically allowed. The geometry does the rest. Equations, if wanted, come out as outputs.

A ball in a bowl could, in principle, be anywhere in the room. Physics allows it only on the bowl's surface. Every physical system is like this. Its constraints confine it to a curved surface, a manifold, inside a vast space of conceivable states. The PDEs describe motion on that surface. **The surface itself is the more fundamental object.** If you can learn it directly, you do not need to derive the PDEs first.

MW makes that move with three deliberate steps.

Step 1 · Describe physics as priors

Physical knowledge is written as probability distributions, chosen on purpose to carry the physics. This is a specification language, not a regulariser.

Physical fact	Prior that carries it	Domain
A fraction lies between 0 and 1	SoC ~ Beta(α, β)	Battery
A quantity cannot exceed a limit	Truncated Normal, $I \le I_{max}$	Battery, RF
A rate constant is positive and spans orders of magnitude	LogNormal (activation energy, SEI, stress coefficients)	Battery, chemistry
A mechanism is either on or off	Bernoulli regime switch (plating, particle cracking)	Battery
Ageing only moves one way	Non-increasing path; forward-only time axis	Battery, world models
Signal power falls with distance	Path-loss and time-of-arrival priors per band	RF spectrum
Measurement is scale-free at the Planck scale	Student-t with ν = 0.8 (no finite moments)	Quantum gravity
States must be orthogonal	Gluing constraint $\langle\psi_i\|\psi_j\rangle = 0$	Quantum chemistry

Step 2 · Cut the hypercube to a manifold

Together, the priors concentrate probability on a thin, curved set of physically realisable states. A Bayesian variational autoencoder, written in Pyro on PyTorch, learns that set. Its decoder induces a Riemannian metric $G(z) = J^T W J$ on the latent space. This is the **state-space reduction** at the heart of MW: the computation leaves the exponential hypercube and moves onto a manifold whose dimension tracks the physics.

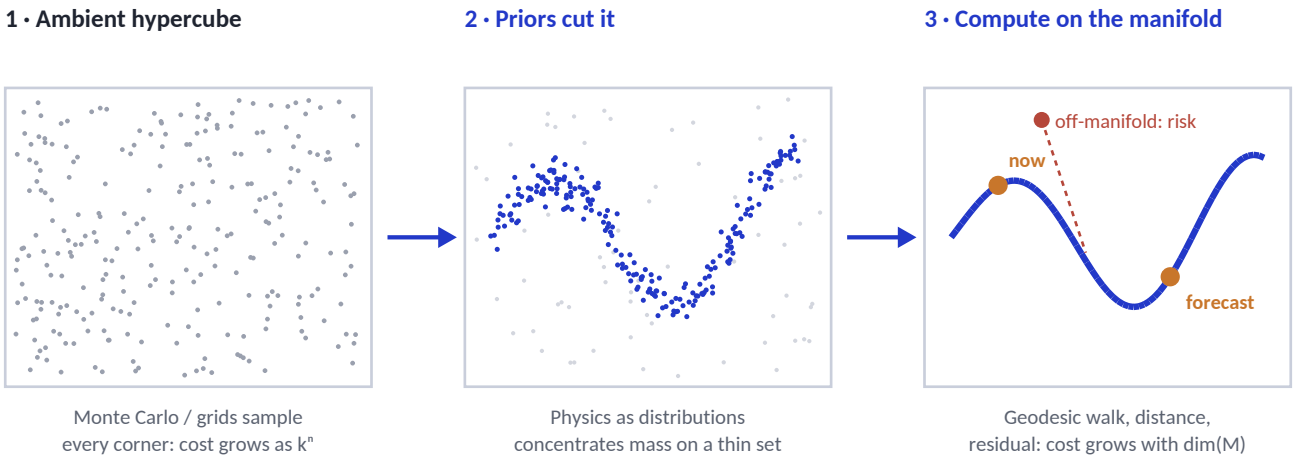


Figure 2. The core move. Left: classic sampling spends effort everywhere. Centre: physics priors concentrate mass on the realisable set. Right: inference, forecasting and risk scoring happen on the learned manifold. Schematic.

Step 3 · Compute on the manifold

Once the manifold exists, the questions physics asks become geometric operations on it:

- **Residuals** per prior tell you which assumption the data rejected, and by how much. A 0.41 residual on monotone degradation, against 0.02 and 0.03 on the others, points straight at charge regeneration: lithium plating or cell reversal.
- **MW distance** from a state to the manifold measures physical implausibility. It is an anomaly score grounded in physics, not in examples of past attacks or failures.
- **Walks** along the manifold produce forecasts without integrating any equation. No step size, no stiffness, no re-solve from scratch.
- **Curvature** carries meaning. Where a battery's ageing regime switches, the metric stretches and the manifold folds; the walk follows the fold, and its projection is the capacity knee.
- **Equations** can be read back off the learned surface as outputs. From pre-1915 orbital data, with the formula absent from the candidate library, MW recovered the 6π coefficient of Einstein's perihelion advance to 0.07 % error.

> Two modes of use. **Mode 1, go fast on known physics:** train once on existing simulations, then query in milliseconds on a CPU. A rotating-black-hole waveform that takes weeks on a supercomputer returns in under a second at 98.5 % frequency-domain match. **Mode 2, explore new physics:** encode a theory as priors, train on real data and read the residuals. Hypothesis testing, done geometrically, without writing a PDE.

3 · How much the state space shrinks

The reduction is not a compression trick applied after the physics. It is the physics. A physically realisable state lives on a set far smaller than the space of conceivable states, and MW computes only there.

System	Ambient space	MW manifold	Consequence
n-qubit register	2^n complex amplitudes	Product of n Bloch spheres with fibre structure: 2n dimensions	1,000 qubits on a laptop in about 2.8 GB of RAM, where a state vector would need ~10^{302} GB
LFP battery pack	Every combination of 16 telemetry channels per cell over life	32-D latent manifold under ageing priors	Life forecast with credible window from one 12–24 h CAN log
RF spectrum	25–40 bands, each with power, timing and spectral shape	Four physics sub-manifolds joined as a fibre bundle	All bands monitored as one geometric object
Quantum-spacetime	Unbounded field configurations	20-D manifold	GR and QM constraints held by one object
Kohn–Sham DFT ($LiFePO_4$)	Full orbital basis	Fibre-bundle cascade of orbital sub-manifolds	0.001 eV MAE on a laptop, without HPC

Figures as reported in the MW engineering documentation. The qubit result applies to physically preparable, structured states; Haar-random states fill the whole Hilbert space and are flagged out of scope by design (Section 9).

4 · What exactly is new

MW is built from established mathematics: Riemannian geometry, variational autoencoders and Bayesian priors. None of these is new. What is new is how they are combined and what they are used for. Physical law, in any domain where a tractable manifold exists, is treated as a geometric object learned from priors and data, and every question is then answered on that object.

What makes it a paradigm rather than a technique is that it ports. The battery engine, the RF sentinel, the qubit emulator and the DFT eigensolver share the identical encode, reparameterise, decode and distance skeleton. Their sub-models differ only in the priors they carry. A domain-specific trick does not travel like that. A way of seeing does.

Each ingredient has a named lineage, and naming it makes the claim stronger. Physics-informed learning (PINNs, neural operators), Bayesian calibration (Kennedy–O'Hagan), generative models of physical states (Boltzmann generators, normalising flows), tensor networks that stitch local state spaces, and the latent-space curvature of Arvanitidis et al. all exist. **What did not exist, to our knowledge, is the conjunction**: priors used deliberately as the primary description of physics, so that the state space collapses to a learned Riemannian manifold on which all modelling, forecasting and risk scoring take place, with one distance serving as both law and risk, carried unchanged from electrochemistry to spacetime.

5 · The algebra and the universal risk measure

A four-operation algebra

MW builds every manifold from four operations. Just as {H, CNOT, T} is universal for quantum circuits, this set is claimed universal for the problems where a tractable manifold exists.

Operation	What it does	Example
Product	Independent sub-manifolds side by side	The first qubit or orbital, unconditioned
FibreBundle	Each component conditioned on the latents of the ones before it	Qubit chains; RF sub-manifolds; battery FibreBundle VAE
GluingMap	Constraints stitching sub-manifolds together	Orthogonality $\langle\psi_i|\psi_j\rangle = 0$ in place of Pauli exclusion
Marginalise	Integrate out sub-manifolds to query a subsystem	One cell of a pack; one band of a spectrum

One object, two structures

The deepest point is not computational reuse. It is that one geometric object carries two universal structures at once: a learned manifold that *is* the physical law, and a single distance on it that *is* the risk or validity measure.

$$\Omega_{MW}(x, M) = \tfrac{1}{2} \Sigma_\alpha \Phi_\alpha \left(x_\alpha - D_\alpha(E(x)) \right)^2$$

- **On the manifold**, the second-order expansion of Ω_{MW} on the pull-back metric $g = J^T W J$ coincides with Synge's world function: half the squared geodesic distance. A Van Vleck determinant $\Delta_{MW} = \det(J_E{}^T H_\Omega J_E)$ supplies an uncertainty bound.
- **Off the manifold**, Synge's construction has no object; it assumes both endpoints lie on the manifold. Ω_{MW} measures extrinsic distance weighted by physics-derived importances Φ_α. This is a genuine extension of the world function, and it is exactly what risk needs: risk is how far a state sits from the manifold of physically valid states.
- **It computes what was out of reach.** In more than a century, closed-form world functions have been found for roughly twenty spacetimes and Van Vleck determinants for a handful. MW computes both approximately, in milliseconds, on any manifold it can learn. The RF sentinel uses the Van Vleck Hessian for passive, GPS-free geolocation.

> The claim, stated precisely. MW does not claim to have invented anomaly detection. It claims the first framework, to our knowledge, in which the anomaly score is (a) a Riemannian distance in a physics-derived metric, (b) the Synge world function when restricted to the learned manifold, with a Van Vleck uncertainty, and (c) transferable across physics without re-derivation.

6 · Combining physics: submanifolds on one manifold

In the PDE route, combining two bodies of physics means reconciling two systems of equations. When their mathematical structures disagree, the reconciliation can take generations. General relativity needs a smooth, differentiable manifold. Quantum mechanics needs probabilistic, non-smooth behaviour at the Planck scale. For more than a hundred years no one has written a single set of equations that respects both.

In MW the problem changes shape. Each body of physics is a set of priors, and each set of priors carves out a submanifold of physically admissible states in a shared ambient space. **Combining two theories means putting both prior sets on one model and learning the manifold that satisfies them together: their intersection.** Nothing has to be derived, and the two theories never have to be rewritten in each other's language. The intersection cannot be found in closed form. It can be learned.

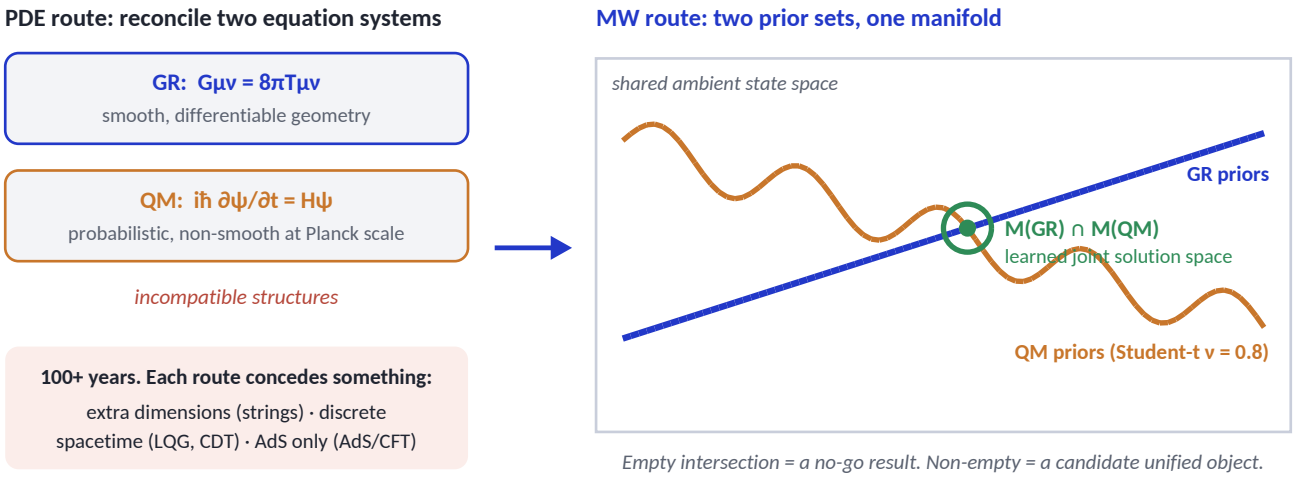


Figure 3. Two ways to combine physics. Left: reconciling two equation systems. Right: two prior sets on one model, with the learned intersection as the joint solution space. Schematic.

The quantum-gravity result

The April 2026 preprint does exactly this. Six physics constraints go onto one Bayesian VAE with a 20-dimensional latent space: Einstein's field equations, Schrödinger evolution, the uncertainty relation, the energy conditions, the (−, +, +, +) metric signature, and quantum measurement encoded as a Student-t prior with ν = 0.8. Only that last prior departs from standard Riemannian geometry. With ν < 1 it has no finite mean or variance and a power-law tail, the mathematical signature of Wheeler's quantum foam. The model learns one manifold that is reported to show five properties together, none of them training objectives: Einstein's equations to 3.3 % residual; unsupervised separation of black-hole, wormhole and foam geometries; Hawking temperature falling as 1/M; a 98.5 % frequency-domain match to LIGO templates; and scale-invariant Planck-scale structure.

Approach	Extra dimensions	Discretises spacetime	AdS only	Single object
String theory	Yes	No	Partial	No
Loop quantum gravity	No	Yes	No	No
Causal dynamical triangulations	No	Yes	No	No
AdS/CFT	Yes	No	Yes	No
MW (Student-t ν = 0.8)	**No**	**No**	**No**	**Yes**

Table 1 of the quantum-gravity preprint: structural requirements of quantum-gravity approaches.

That is the structural claim, and it stands on its own: **MW is the first single probabilistic geometric object that holds the GR and QM constraint sets together without adding dimensions, discretising spacetime or restricting to Anti-de Sitter geometry.** The preprint is careful not to claim quantum gravity solved. Its training data are synthetic and ν = 0.8 is determined empirically. It states a falsifiable prediction that anyone can run on the open code: replace the Student-t with any finite-variance prior (ν > 1) and the foam cluster should vanish while the GR

sector survives.

A machine for testing unification

The same move works for any pair of theories. Point MW at any collection of constraint sets and read off whether their intersection is empty and, if not, what geometry it has. Every outcome is a result. A non-empty intersection is a candidate unified object. An empty one is a no-go theorem for that family of priors. And the method is already in daily use at smaller scale:

- **Batteries:** electrochemistry, heat (Arrhenius, Joule) and charge accounting (coulomb closure) are separate physics, combined as priors on one manifold. A cyber-attack is a state that satisfies the protocol but not all three at once.
- **RF spectrum:** four physics sub-manifolds joined as one fibre bundle, so 40 bands are read as a single object.
- **Quantum chemistry:** orbital sub-manifolds glued by orthogonality, replacing Pauli exclusion with a gluing map.

The algebra of Section 5 is what makes this routine: Product places sub-manifolds side by side, FibreBundle conditions one on another, GluingMap imposes the shared constraints, and Marginalise reads out a single theory again.

7 · One engine, eight domains

The breadth below is not scattered effort. It is the evidence that the reframe is real: the same construction, with only the priors changed, does useful work from electrochemistry to the electromagnetic spectrum to spacetime.

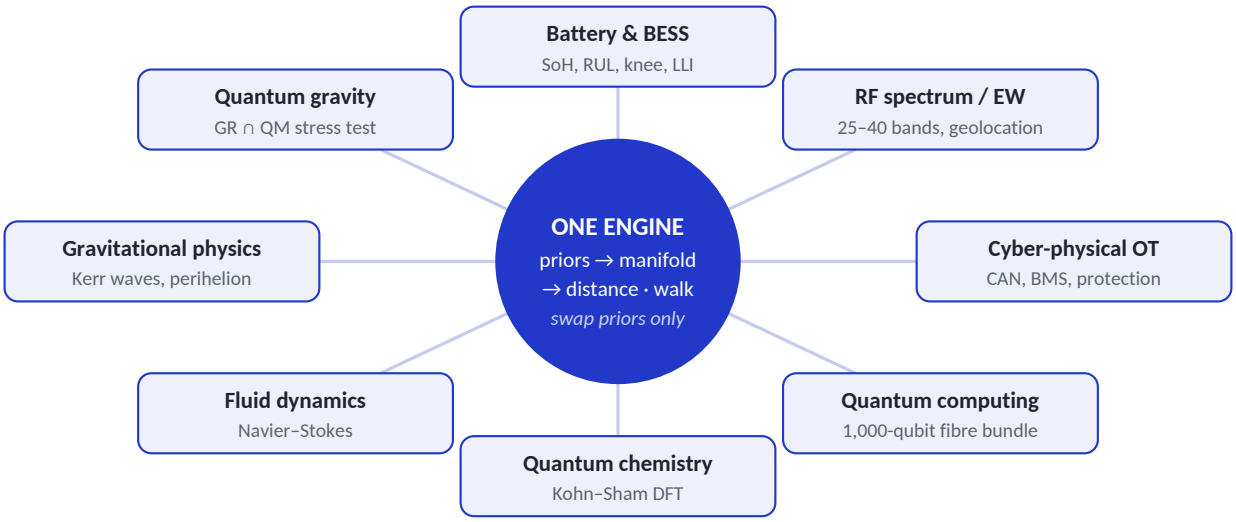


Figure 4. The MW engine and the domains it has been carried into.

7.1 · Battery analytics and BESS (BayesianBESS, MW Battery Engine)

Priors	Arrhenius temperature acceleration, √t SEI growth, cycle, C-rate, SoC and DoD stress, plating, swelling, particle cracking; SoH ≤ 1 and no recovery.
What it does	Reads an ordinary 12–24 h BMS/CAN log and returns per-cell resistance, lithium-inventory loss from dQ/dV peak shift, a life forecast with a 90 % credible window, P(end of life before N cycles), lever ranking against posterior σ, and ten physics integrity laws.
Why a short log works	It is Kennedy–O'Hagan calibration with physical priors in place of Gaussian processes: the non-linear family of ageing paths comes from physics and a synthetic fleet; the log locates the cell on it.
Evidence	95 public LFP cells (Sandia, HUST), held out: 2.4–3.4 pp median RMSE against 18–21 pp for the physics formula alone. Out of the training domain, 3.0 pp where an unconstrained model reached 20.4 pp. A published SEI-cracking study reproduced on the author's own data, confirmed by the author.

7.2 · RF spectrum and cognitive electronic warfare (MW RF Sentinel)

Priors	Propagation physics per band: path loss, time of arrival, expected spectral occupancy. No signature database.
What it does	Monitors 25 bands, extended to 40 across four physics sub-manifolds joined as a fibre bundle, as a single geometric object. Flags departures from physics-predicted norms, classifies multi-drone swarms, and geolocates emitters passively with the Van Vleck Hessian, without GPS.
Field result	On 20 March 2026 a single Rs 5,800 RTL-SDR node in Thane detected a real GNSS suppression event at about 9,400 times the 72-hour baseline (9,393×). It used no signature database, and no threshold-based system nearby reported it. This is an observed field event, not a simulation.
Defence submission	Proposed to the Ministry of Defence's iDEX **ADITI 4.0** programme under **Problem Statement 5, Cognitive Electronic Warfare System (Indian Army)**, as the MW Cognitive EW System: a wholly indigenous intelligence layer that reasons from physical laws rather than stored signatures, so it detects unknown threats as readily as known ones and is robust to adversary firmware changes.
Test bed	A 45-scenario red-team war-game, scored band by band against ground truth, and a forked world model in which MW acts as the physics validator for RL policies.

7.3 · Cyber-physical and OT security

Idea	Industrial buses such as CAN have no authentication, but physics is hard to forge. A spoofed state of charge must still match the current that flowed; faked temperatures must still follow Joule heating; manipulated voltages must still relax correctly.
What it does	Catches commands that are protocol-valid but physically impossible, and quietly disabled protection, as departures from the manifold. Deployed in the BESS Cyber-Physical Red Team Range and the OT security stack for utilities.

7.4 · Quantum computing and quantum chemistry (MWKit)

Quantum computing	A 1,000-qubit emulator on a consumer laptop, working on a 2n-dimensional fibre bundle of Bloch spheres instead of the 2^n state vector. Haar-random states score maximum MW distance and separate cleanly from physical states.
Quantum chemistry	A Kohn–Sham DFT eigensolver for $LiFePO_4$ built as a fibre-bundle cascade of orbital sub-manifolds, with orthogonality as a gluing map, at 0.001 eV mean absolute error on a laptop. Where the manifold is unsure (high σ), it defers to VASP.

7.5 · Fluid dynamics, gravitational physics and quantum gravity

Navier–Stokes	Laminar and coherent flow lives on the learned flow manifold; vortex cores and discontinuities are located and flagged rather than smoothed over. Wind-turbine design engineers are being trained to use MW for 3-D flow work.
Gravitational physics	Kerr black-hole waveforms in under a second on a CPU at 98.5 % frequency-domain match to LIGO templates. Recovery of the perihelion formula's 6π coefficient from pre-1915 data.
Quantum gravity (stress test)	A Student-t(ν = 0.8) prior on the measurement process yields a 20-D manifold reported to hold five properties together: Einstein's equations to 3.3 % residual, unsupervised separation of black-hole, wormhole and foam geometries, Hawking temperature falling as 1/M, a 98.5 % LIGO match, and scale-invariant Planck-scale structure. GR and QM priors held on one manifold (Section 6).

8 · A fourth family of world models

World models learn an internal simulator so an agent can plan inside it. The field has three families: latent imagination (Dreamer, TD-MPC2), joint-embedding prediction (JEPA) and generative video (Genie, Cosmos). In all three, time is an index the model steps along. Nothing stops an imagined rollout from taking a battery from 70 % to 95 % health.

MW gives a fourth family. The model is a manifold of admissible states with an explicit time axis that only moves forward. A forecast is a walk on the manifold, not a rollout of a learned transition. The Lorentzian extension gives the time axis a negative sign, as in the spacetime metric (−, +, +, +), so every state carries a causal cone: a transition that changes too much for the time elapsed lies outside it and is inadmissible. For the first time, a world model can express "this state cannot be reached from that one."

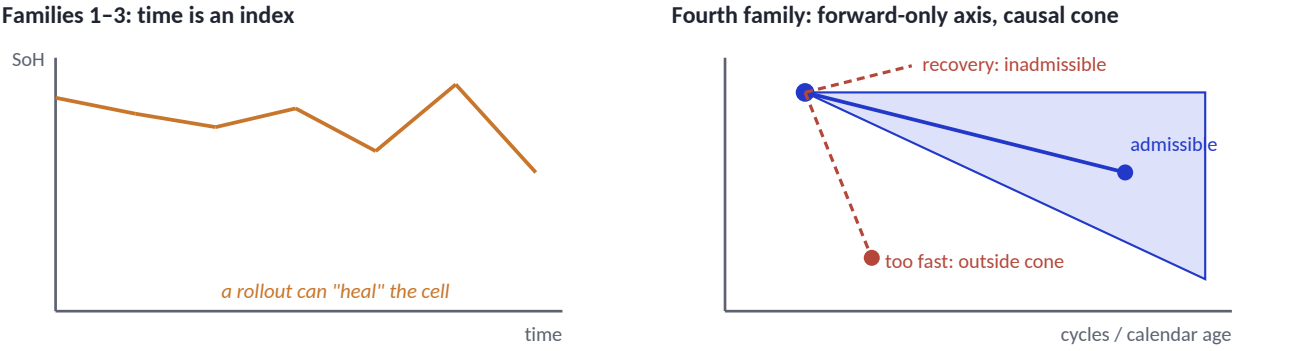


Figure 5. Left: a learned transition function can "heal" a cell. Right: a forward-only axis rules out recovery, and the Lorentzian cone rules out change faster than physics allows. Schematic.

The same idea positions MW as a **physics validator for any world model**: wrapped around a learned simulator, the manifold rejects imagined states that physics forbids. It is already in production for EV fleet battery analytics in place of a Doyle–Fuller–Newman model, and wired into the RF war-game.

9 · The boundary is part of the contribution

A foundational claim states its boundary precisely. MW carries its own boundary inside the code, as a second mode.

Domain	Mode A · manifold exists	Mode B · physics leaves the manifold	How the boundary is drawn
Battery / BESS	Normal ageing and protection behaviour	Silent protection failure, sensor fault, cyber-attack	Integrity laws fire; MW distance > 3σ
RF spectrum	Physics-predicted band occupancy	Jamming, spoofing, unknown emitters	Reconstruction ratio far above baseline
Quantum computing	Basis and structured superpositions	Haar-random states; Shor and Grover	Maximum MW distance; flagged out of scope
Quantum chemistry	Orbital manifold the model trusts	High-σ structures	Defer to VASP
Navier–Stokes	Laminar and coherent flow	Vortex cores, discontinuities, turbulence	Located and flagged, not smoothed

The states with no low-dimensional manifold, such as Haar-random quantum states or fully developed turbulence, are the states nature does not prepare and nobody needs to compute. So the framework's domain is the physically meaningful state spaces, and its edge is the edge nature draws. Where a manifold exists, MW solves on it. Where it does not, MW detects, locates and puts an uncertainty bar on the departure. Both are results.

MW also states plainly what it does not claim. It does not claim to have solved quantum gravity; the training data there are synthetic and $\nu = 0.8$ is empirical. It does not claim Navier–Stokes existence and smoothness; it locates singularities. A battery life forecast is not a safety system and cannot see thermal runaway. Uncertainty bands on open battery data still run narrower than they claim, and that recalibration is under way.

10 · How the claim is being earned

MW is being validated domain by domain, on data and events the authors did not generate.

Result	Source of truth	Status
Sandia + HUST, 95 LFP cells, held-out SoH forecasts	Open public datasets	Complete, reproducible with one command
SEI-cracking study reproduced	External author's data; author confirmed	Complete
Thane GNSS suppression event, 9,393×	Live RF environment, real event	Observed in the field
Independent reproduction of MW results	Outside researchers	Reported
Perihelion 6π coefficient from pre-1915 data	Historical astronomical data	Complete
Kerr waveforms, 98.5 % LIGO match	LIGO template bank	Complete (synthetic generation)
Quantum-gravity five properties	Synthetic constraint data	Open to replication, with a falsifiable test
Battery, RF and fluids on independent data	Partner and open data	Target: early 2027

Next steps that harden the foundation

- **The on/off-manifold lemma** as a two-page result for geometers: the pull-back metric, second-order coincidence with Synge's world function, the Van Vleck Hessian and the off-manifold extension.
- **A blind prospective battery study**: life intervals filed before cells reach end of life, scored for coverage.
- **The ν > 1 falsification** of the quantum-gravity result: any finite-variance prior should erase the foam cluster while keeping the GR sector. Anyone with the open code can run it.
- **An open code companion** (MW_Geometric_Intelligence) so that every claim can be checked without trusting the authors.

11 · The contribution in one paragraph

> *The Modak–Walawalkar Framework replaces the 339-year practice of deriving and solving governing equations with a different object. Physics is described deliberately as Bayesian priors. The priors cut the ambient state space down to the thin set of physically realisable states, which a Bayesian variational autoencoder learns as a Riemannian manifold. On that manifold, one distance is both the physical model and the measure of risk: Synge's world function on it, a physics-grounded extension off it. Four operations, Product, FibreBundle, GluingMap and Marginalise, build the manifold for any domain where one exists, and the same engine runs, with only its priors changed, across batteries, RF spectrum and electronic warfare, cyber-physical security, quantum computing, quantum chemistry, fluid dynamics, gravitational physics and quantum gravity. Different bodies of physics combine by placing their priors on one manifold and learning the intersection, which is how GR and QM are held by one object. Given a forward-only, Lorentzian time axis, it becomes a fourth family of world models in which physically impossible futures are inadmissible. Think manifolds, not PDEs.*

How the work developed

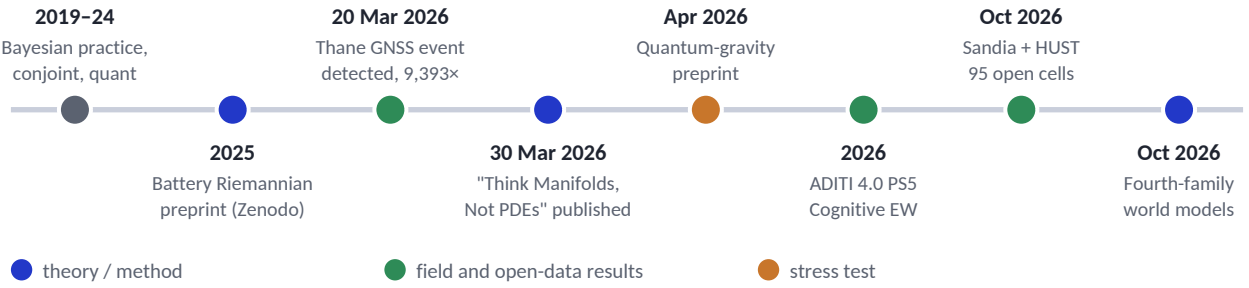


Figure 6. Milestones, 2019–2026.

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